Generalized sleep decoding with basal ganglia signals in multiple movement disorders

Paper: Yin et al. (2024). Human sleep-decoding study.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

What implanted electrodes can measure, and how feedback could account for sleep. Evaluates a sleep-state estimator across people, diagnoses and recording targets.

How does it answer the general questions?

Measured. Synchronized polysomnography and intracranial field potentials provided training labels and neural features. After exclusions, 140 nights from 121 patients were analysed. BGOOSE achieved approximately 85% average accuracy for awake/NREM/REM classification; additional ECoG improved decoding (PDF pp. 1–7).

Usable insight. Cross-subject evaluation and external datasets support using neural sensing to estimate context without individual polysomnography calibration. Performance is not uniform across stages: separating NREM substages is substantially harder. Classification results do not themselves show that sleep-triggered stimulation improves sleep (PDF pp. 6–8, 11).

Model relevance. A sleep estimator could select operating conditions or feedback policies. Its spectral, waveform and entropy features require an observation/processing layer beyond a linear $y = Cx$. No stimulation-to-neural-state dynamics or delayed-model coefficients are identified. Sleep quality would require its own validated performance output.

What papers does it cite that could provide more information?

- Smyth et al. (2023), *Adaptive deep brain stimulation for sleep stage targeting in Parkinson's disease*: intervention-oriented follow-up (ref. 12, PDF p. 12).
- Gilron et al. (2021), *Long-term wireless streaming of neural recordings for circuit discovery and adaptive stimulation in individuals with Parkinson's disease*: chronic sensing (ref. 28, PDF p. 12).